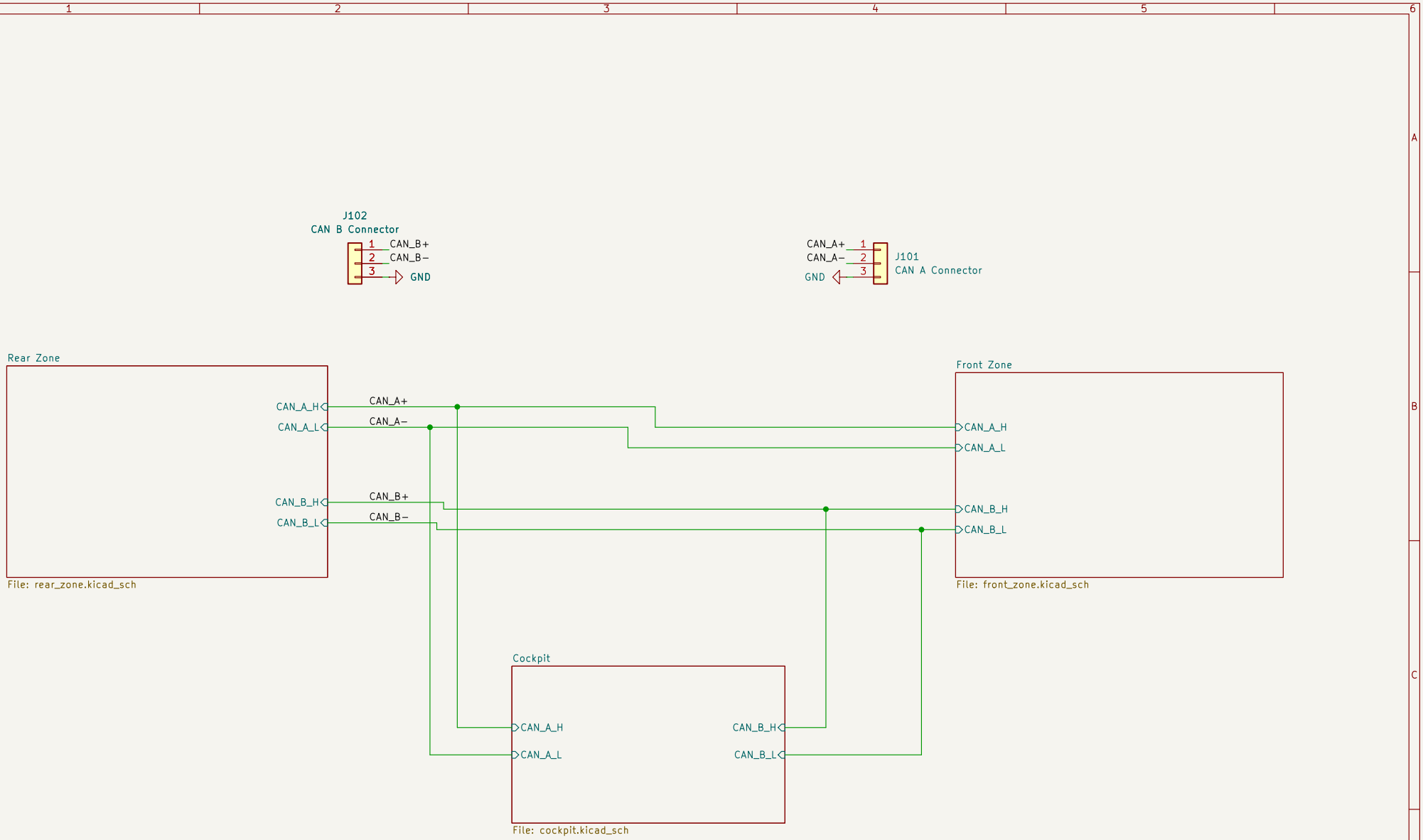
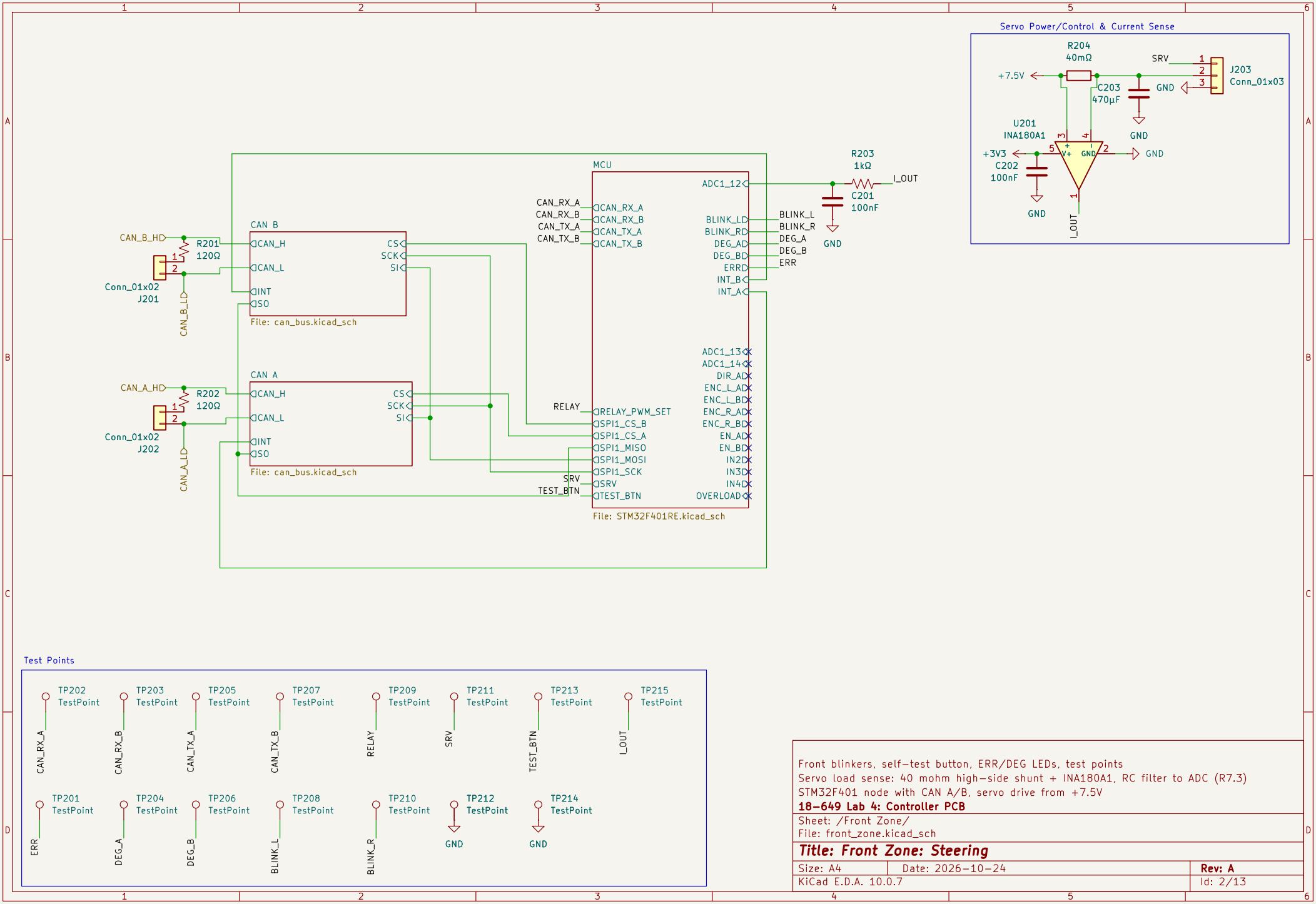
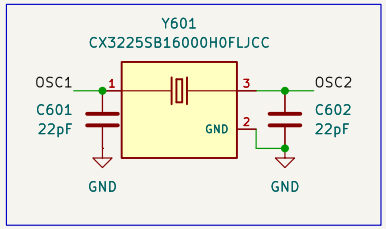
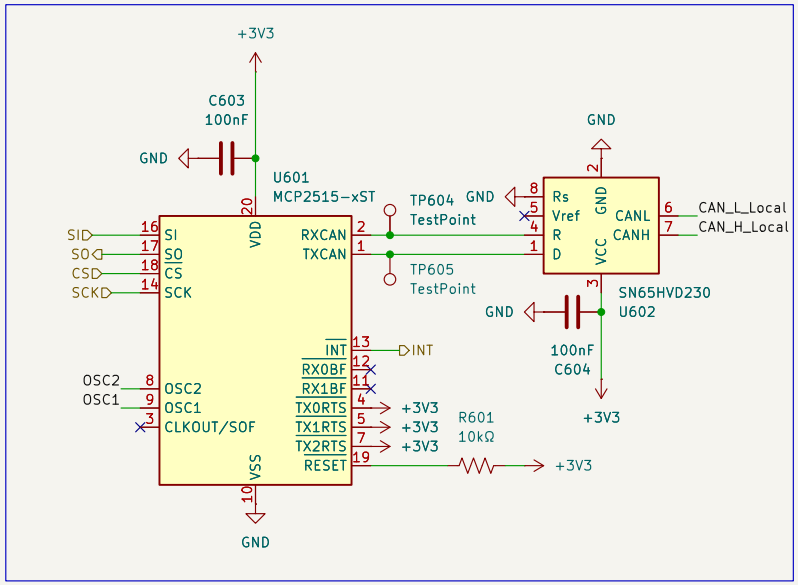
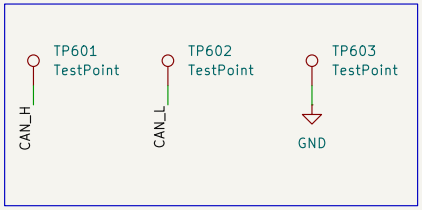
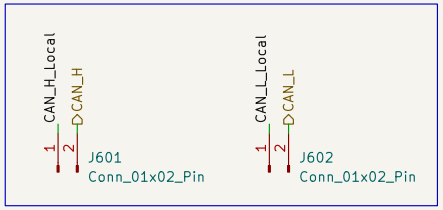

J102
CAN B Connector
1 CAN_B+
2 CAN_B−
3 GND
CAN_A+ 1
CAN_A− 2
GND 3
J101
CAN A Connector
Rear Zone
CAN_A_H
CAN_A_L
CAN_B_H
CAN_B_L
CAN_A+
CAN_A−
CAN_B+
CAN_B−
File: rear_zone.kicad_sch
Front Zone
CAN_A_H
CAN_A_L
CAN_B_H
CAN_B_L
File: front_zone.kicad_sch
Cockpit
CAN_A_H
CAN_A_L
CAN_B_H
CAN_B_L
File: cockpit.kicad_sch
Power
File: power.kicad_sch
CAN A and CAN B: separate linear buses (CAN_A+/−, CAN_B+/−), 120 ohm termination at each end
Front (steering), Rear (drivetrain) and Cockpit zones, plus Power
18−649 Lab 4: Controller PCB
Sheet: /
File: front_rear_dual_can.kicad_sch
Title: Controller PCB: Top Level
Size: A4
Date: 2026−10−24
Rev: A
KiCad E.D.A. 10.0.7
Id: 1/13


Servo Power/Control & Current Sense
R204
40mΩ
+7.5V
SRV
J203
Conn_01x03
C203
470µF
GND
U201
INA180A1
+3V3
C202
100nF
V+
GND
I_OUT
R203
1kΩ
C201
100nF
MCU
ADC1_12
CAN_RX_A
CAN_RX_B
CAN_TX_A
CAN_TX_B
BLINK_L
BLINK_R
DEG_A
DEG_B
ERR
INT_B
INT_A
ADC1_13
ADC1_14
DIR_A
ENC_L_A
ENC_L_B
ENC_R_A
ENC_R_B
EN_A
EN_B
IN2
IN3
IN4
OVERLOAD
RELAY
RELAY_PWM_SET
SPI1_CS_B
SPI1_CS_A
SPI1_MISO
SPI1_MOSI
SPI1_SCK
SRV
TEST_BTN
File: STM32F401RE.kicad_sch
CAN B
CAN_B_HD
R201
120Ω
Conn_01x02
J201
CAN_B_LD
CAN_H
CAN_L
INT
SO
CS
SCK
SI
File: can_bus.kicad_sch
CAN A
CAN_A_HD
R202
120Ω
Conn_01x02
J202
CAN_A_LD
File: can_bus.kicad_sch
Test Points
TP202 TestPoint CAN_RX_A
TP203 TestPoint CAN_RX_B
TP205 TestPoint CAN_TX_A
TP207 TestPoint CAN_TX_B
TP209 TestPoint RELAY
TP211 TestPoint SRV
TP213 TestPoint TEST_BTN
TP215 TestPoint I_OUT
TP201 TestPoint ERR
TP204 TestPoint DEG_A
TP206 TestPoint DEG_B
TP208 TestPoint BLINK_L
TP210 TestPoint BLINK_R
TP212 TestPoint GND
TP214 TestPoint GND
Front blinkers, self-test button, ERR/DEG LEDs, test points
Servo load sense: 40 mohm high-side shunt + INA180A1, RC filter to ADC (R7.3)
STM32F401 node with CAN A/B, servo drive from +7.5V
18-649 Lab 4: Controller PCB
Sheet: /Front Zone/
File: front_zone.kicad_sch
Title: Front Zone: Steering
Size: A4
Date: 2026-10-24
Rev: A
KiCad E.D.A. 10.0.7
Id: 2/13









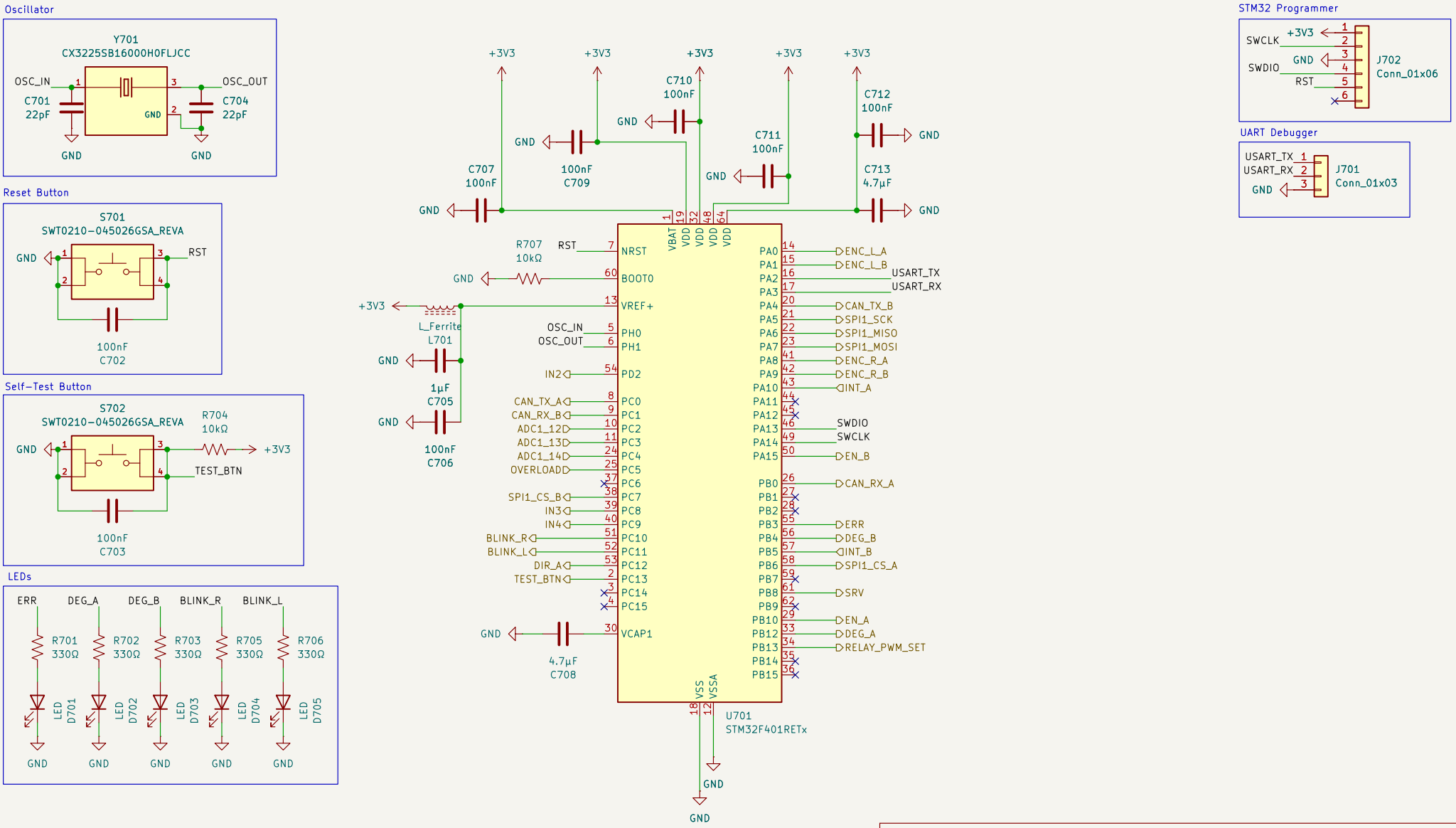
CAN_TX/CAN_RX test points per node
Isolation jumpers break CAN_H and CAN_L (R4.8); CAN_H/CAN_L test header (R4.6)
One per bus per zone (R4.1); own 16 MHz crystal (R4.2)

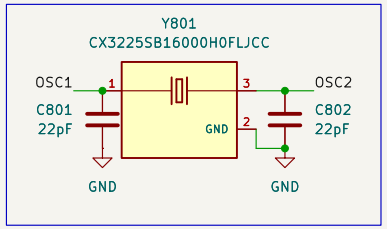
18-649 Lab 4: Controller PCB

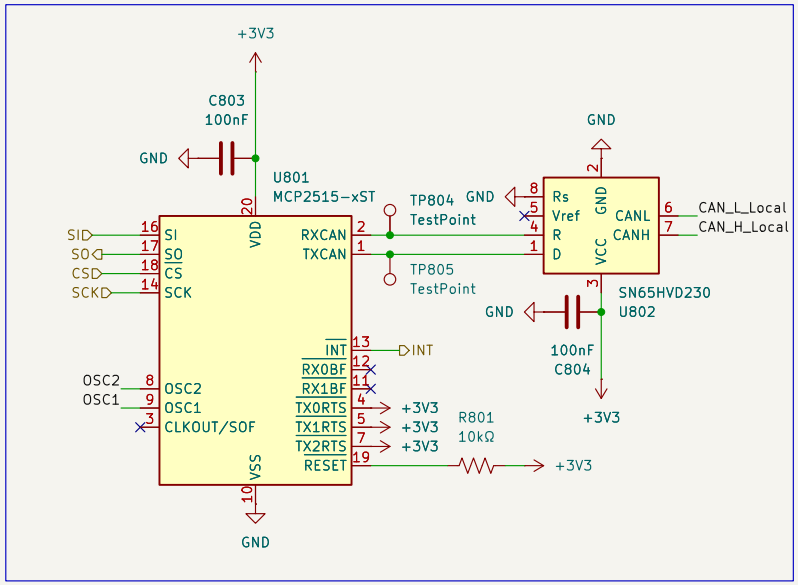
Sheet: /Front Zone/CAN A/
File: can_bus.kicad_sch

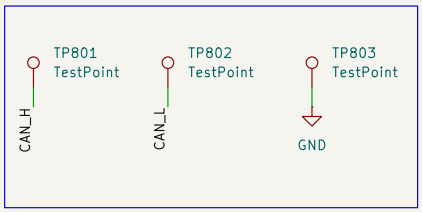
Title: CAN Node: MCP2515 + SN65HVD230 (6 instances)

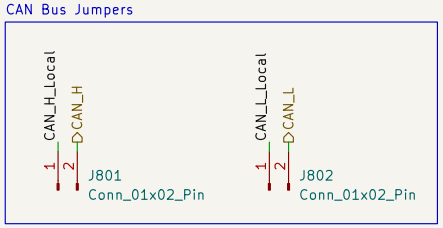
Size: A4	Date: 2026-10-24	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 6/13


Oscillator
Y701
CX3225SB16000H0FLJCC
OSC_IN
OSC_OUT
C701
22pF
C704
22pF
GND
Reset Button
S701
SWT0210-045026GSA_REVA
RST
100nF
C702
Self-Test Button
S702
SWT0210-045026GSA_REVA
R704
10kΩ
+3V3
TEST_BTN
100nF
C703
LEDs
ERR
DEG_A
DEG_B
BLINK_R
BLINK_L
R701 330Ω
R702 330Ω
R703 330Ω
R705 330Ω
R706 330Ω
LED D701
LED D702
LED D703
LED D704
LED D705
C707
100nF
C709
100nF
C710
100nF
C711
100nF
C712
100nF
C713
4.7µF
R707
10kΩ
RST
L_Ferrite
L701
1µF
C705
100nF
C706
4.7µF
C708
U701
STM32F401RETx
NRST
BOOT0
VREF+
PH0
PH1
PD2
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14
PC15
VCAP1
VBAT
VDD
VSS
VSSA
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
ENC_L_A
ENC_L_B
USART_TX
USART_RX
CAN_TX_B
SPI1_SCK
SPI1_MISO
SPI1_MOSI
ENC_R_A
ENC_R_B
INT_A
SWDIO
SWCLK
EN_B
CAN_RX_A
ERR
DEG_B
INT_B
SPI1_CS_A
SRV
EN_A
DEG_A
RELAY_PWM_SET
CAN_TX_A
CAN_RX_B
ADC1_12
ADC1_13
ADC1_14
OVERLOAD
SPI1_CS_B
IN3
IN4
BLINK_R
BLINK_L
DIR_A
TEST_BTN
STM32 Programmer
SWCLK
SWDIO
RST
J702
Conn_01x06
UART Debugger
USART_TX
USART_RX
J701
Conn_01x03
SWD header, NRST, BOOT0, VCAP and VDDA/VREF+ filtering, decoupling
Shared by Front and Rear zones; each zone has its own 16 MHz crystal (R1)
18-649 Lab 4: Controller PCB
Sheet: /Front Zone/MCU/
File: STM32F401RE.kicad_sch
Title: Zone MCU: STM32F401RET6 (Front and Rear)
Size: A4
Date: 2026-10-24
Rev: A
KiCad E.D.A. 10.0.7
Id: 7/13









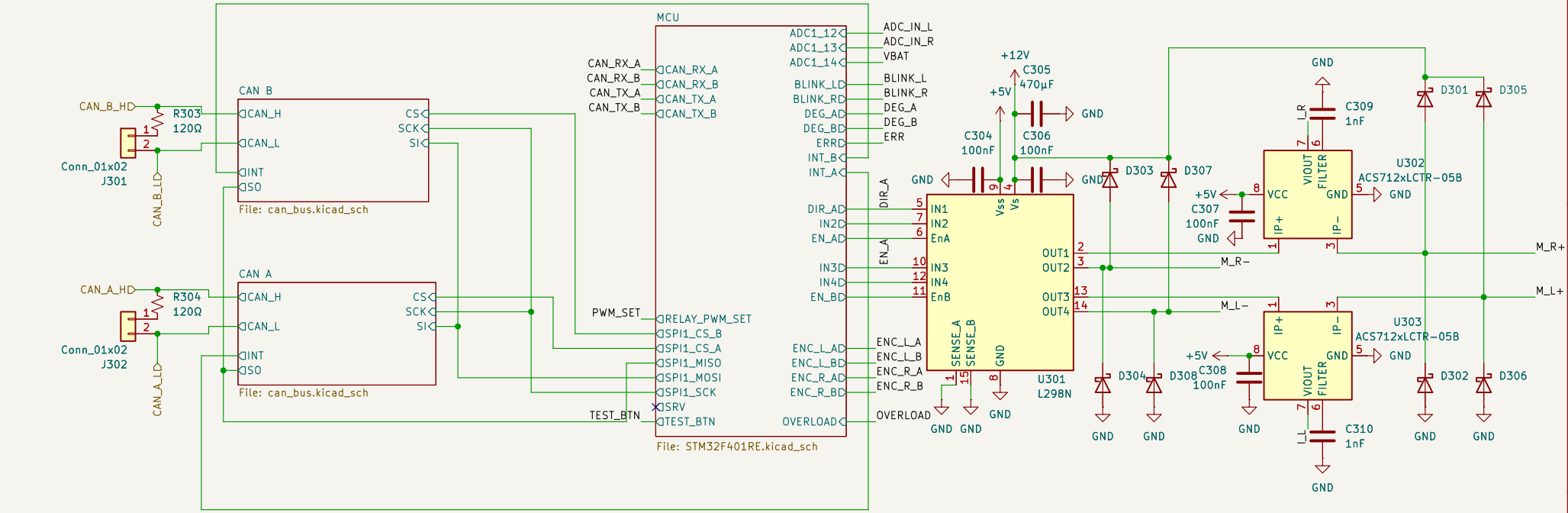
CAN_TX/CAN_RX test points per node
Isolation jumpers break CAN_H and CAN_L (R4.8); CAN_H/CAN_L test header (R4.6)
One per bus per zone (R4.1); own 16 MHz crystal (R4.2)

18-649 Lab 4: Controller PCB

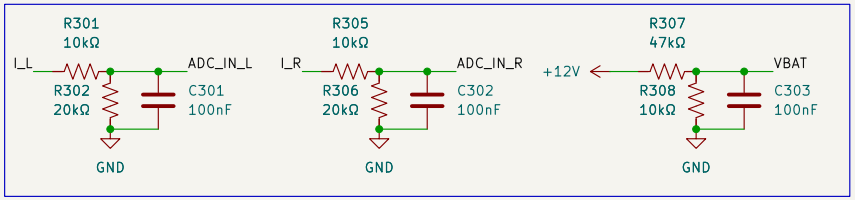
Sheet: /Front Zone/CAN B/
File: can_bus.kicad_sch

Title: CAN Node: MCP2515 + SN65HVD230 (6 instances)

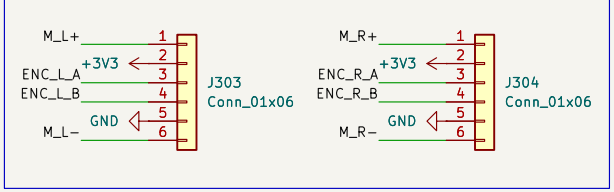
Size: A4	Date: 2026-10-24	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 8/13



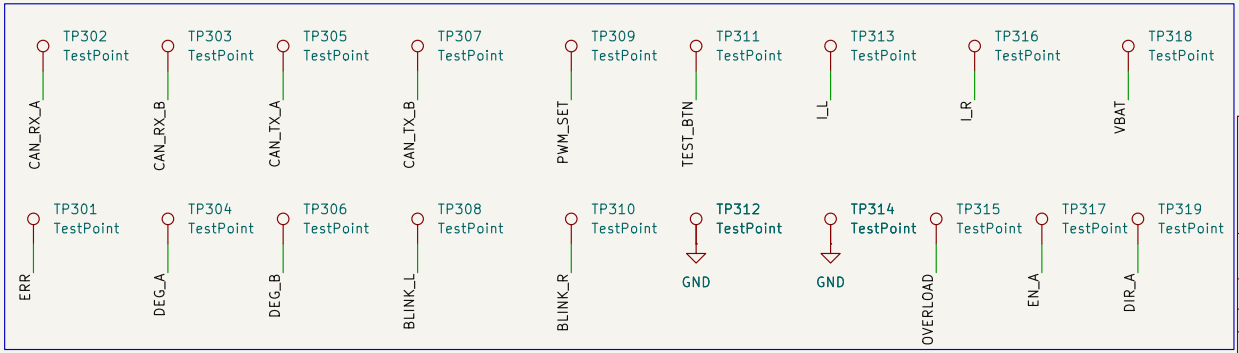
Voltage Dividers



Motor Encoder Connectors



Test Points



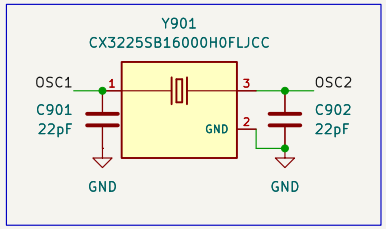
Rear blinkers, self-test button, ERR/DEG LEDs, test points
ACS712 current sense per motor (R6.1, R6.5), encoder inputs (R5.1), VBAT sense
STM32F401 node with CAN A/B, L298N H-bridge with Schottky flyback diodes

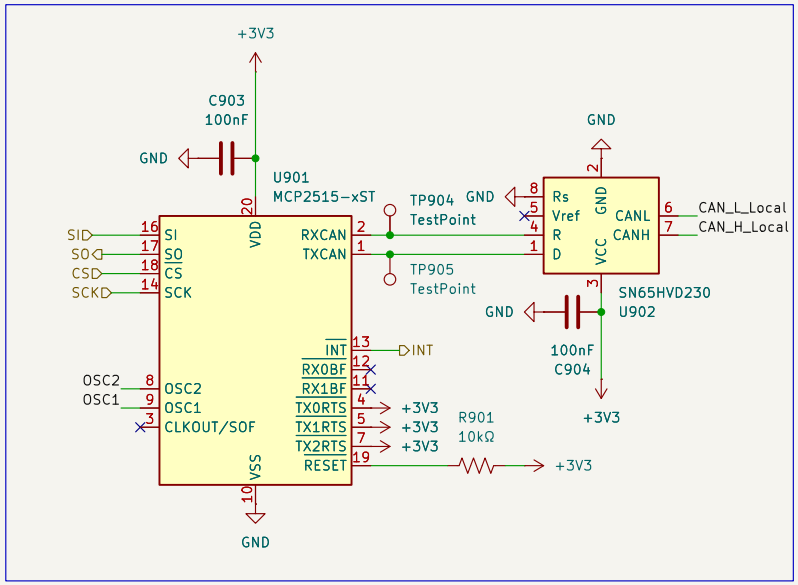
18-649 Lab 4: Controller PCB

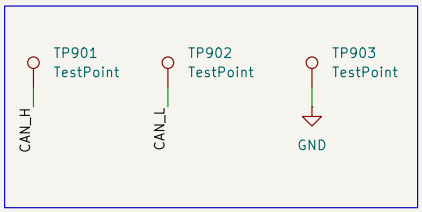
Sheet: /Rear Zone/
File: rear_zone.kicad_sch

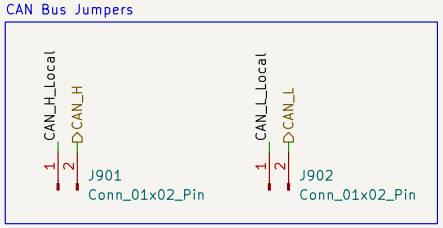
Title: Rear Zone: Drivetrain

Size: A4	Date: 2026-10-24	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 3/13









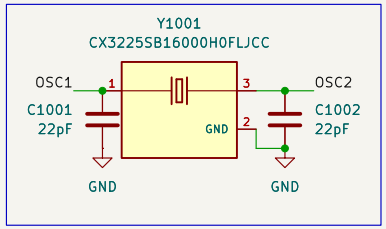
CAN_TX/CAN_RX test points per node
Isolation jumpers break CAN_H and CAN_L (R4.8); CAN_H/CAN_L test header (R4.6)
One per bus per zone (R4.1); own 16 MHz crystal (R4.2)

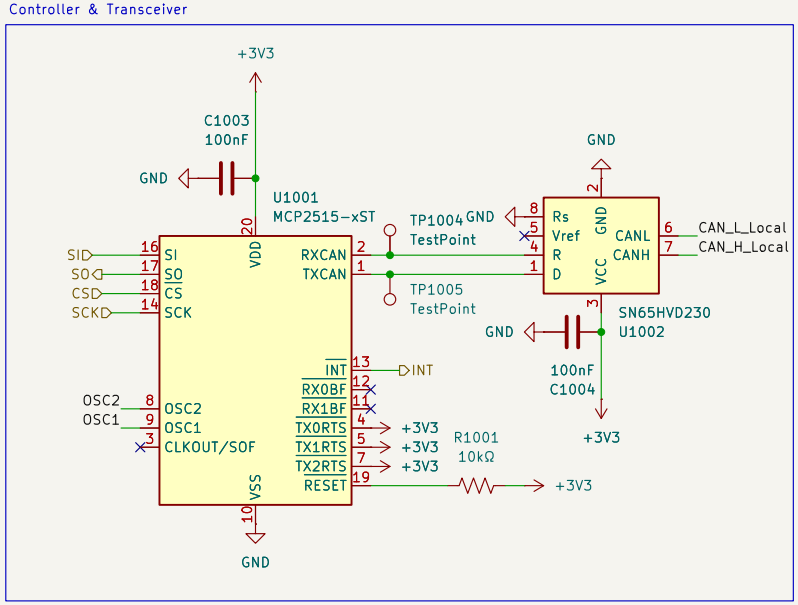
18-649 Lab 4: Controller PCB

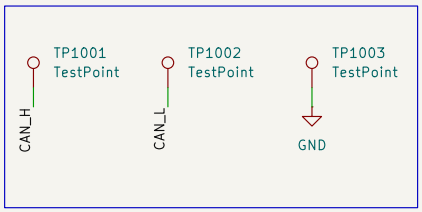
Sheet: /Rear Zone/CAN A/
File: can_bus.kicad_sch

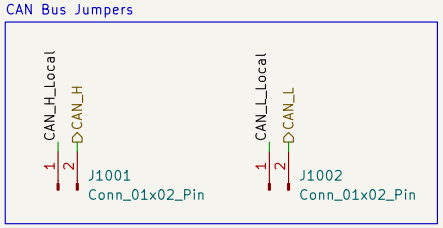
Title: CAN Node: MCP2515 + SN65HVD230 (6 instances)

Size: A4	Date: 2026-10-24	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 9/13









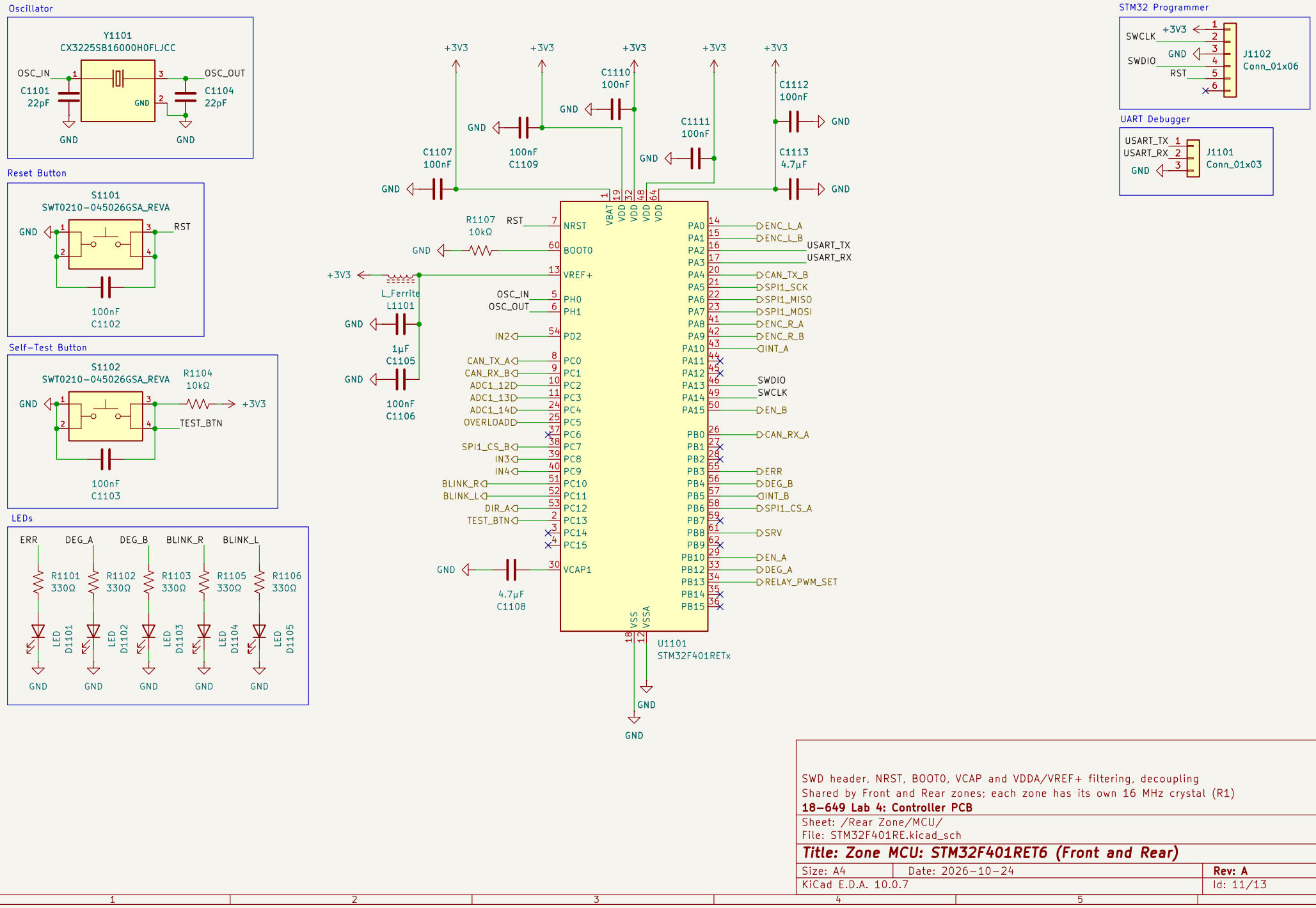
CAN_TX/CAN_RX test points per node
Isolation jumpers break CAN_H and CAN_L (R4.8); CAN_H/CAN_L test header (R4.6)
One per bus per zone (R4.1); own 16 MHz crystal (R4.2)

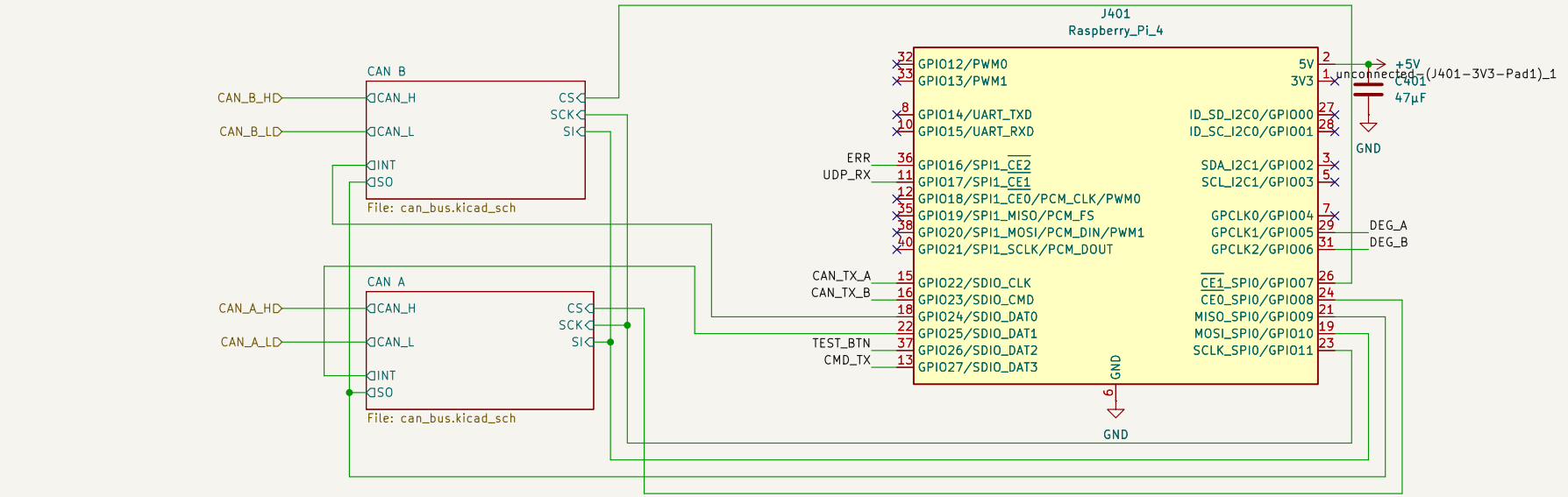
18-649 Lab 4: Controller PCB

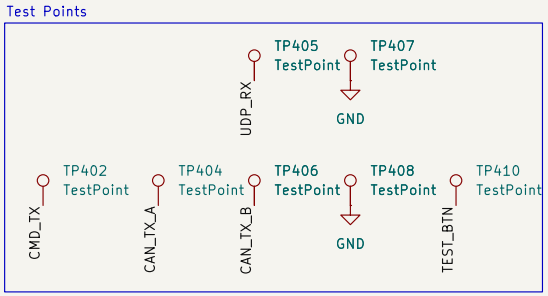
Sheet: /Rear Zone/CAN B/
File: can_bus.kicad_sch

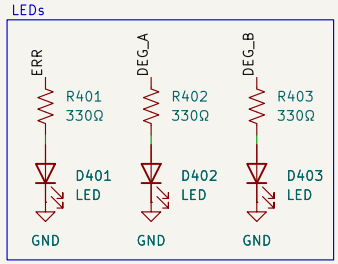
Title: CAN Node: MCP2515 + SN65HVD230 (6 instances)

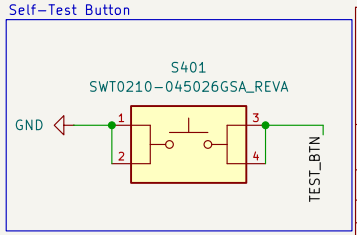
Size: A4	Date: 2026-10-24	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 10/13


Oscillator
Y1101
CX3225SB16000H0FLJCC
OSC_IN
OSC_OUT
C1101
22pF
C1104
22pF
GND
Reset Button
S1101
SWT0210-045026GSA_REVA
RST
100nF
C1102
Self-Test Button
S1102
SWT0210-045026GSA_REVA
R1104
10kΩ
+3V3
TEST_BTN
100nF
C1103
LEDs
ERR
DEG_A
DEG_B
BLINK_R
BLINK_L
R1101 330Ω
R1102 330Ω
R1103 330Ω
R1105 330Ω
R1106 330Ω
LED D1101
LED D1102
LED D1103
LED D1104
LED D1105
C1107 100nF
C1109 100nF
C1110 100nF
C1111 100nF
C1112 100nF
C1113 4.7µF
R1107 10kΩ
L_Ferrite L1101
1µF C1105
100nF C1106
4.7µF C1108
U1101
STM32F401RETx
NRST
BOOT0
VREF+
PH0
PH1
PD2
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14
PC15
VCAP1
VBAT
VDD
VSS
VSSA
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
ENC_L_A
ENC_L_B
USART_TX
USART_RX
CAN_TX_B
SPI1_SCK
SPI1_MISO
SPI1_MOSI
ENC_R_A
ENC_R_B
INT_A
SWDIO
SWCLK
EN_B
CAN_RX_A
ERR
DEG_B
INT_B
SPI1_CS_A
SRV
EN_A
DEG_A
RELAY_PWM_SET
CAN_TX_A
CAN_RX_B
ADC1_12
ADC1_13
ADC1_14
OVERLOAD
SPI1_CS_B
IN3
IN4
BLINK_R
BLINK_L
DIR_A
TEST_BTN
STM32 Programmer
SWCLK
SWDIO
RST
J1102
Conn_01x06
UART Debugger
USART_TX
USART_RX
J1101
Conn_01x03
SWD header, NRST, BOOT0, VCAP and VDDA/VREF+ filtering, decoupling
Shared by Front and Rear zones; each zone has its own 16 MHz crystal (R1)
18-649 Lab 4: Controller PCB
Sheet: /Rear Zone/MCU/
File: STM32F401RE.kicad_sch
Title: Zone MCU: STM32F401RET6 (Front and Rear)
Size: A4
Date: 2026-10-24
Rev: A
KiCad E.D.A. 10.0.7
Id: 11/13









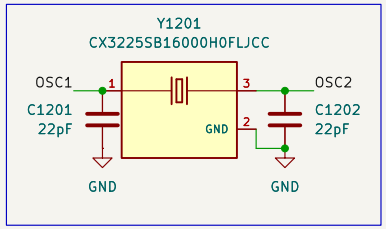
Self-test button, 47 uF bulk on Pi 5 V
Two on-board CAN nodes (A and B), DEG_A/DEG_B/ERR LEDs driven by Pi GPIOs (R4.7)
Pi 40-pin header powered from board +5V; SPI0 with CE0/CE1 and two interrupt lines

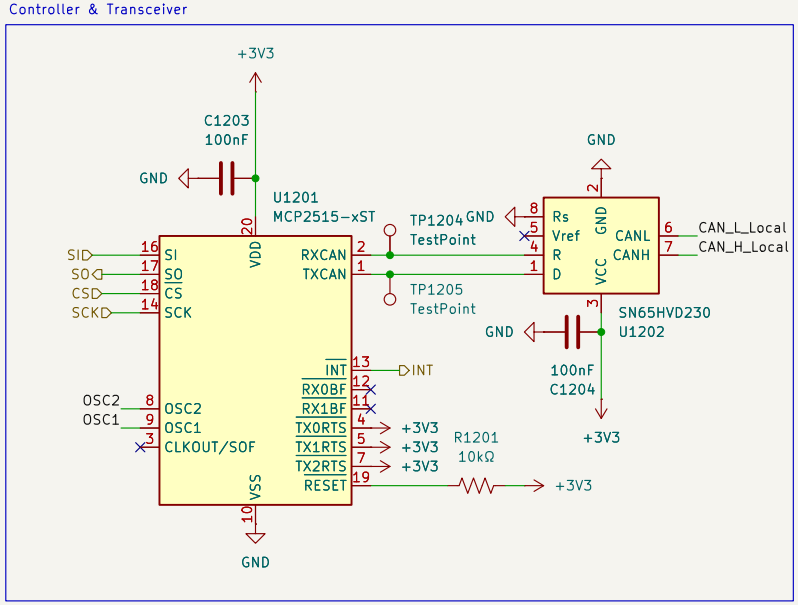
18-649 Lab 4: Controller PCB

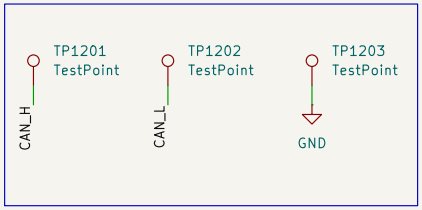
Sheet: /Cockpit/
File: cockpit.kicad_sch

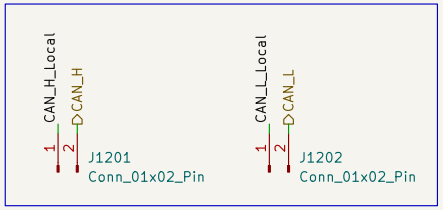
Title: Cockpit Zone: Raspberry Pi 4 Interface

Size: A4	Date: 2026-10-24	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 4/13









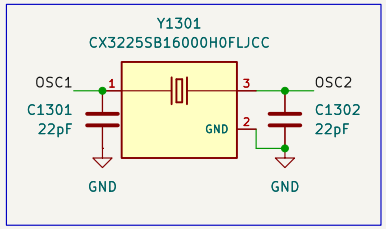
CAN_TX/CAN_RX test points per node
Isolation jumpers break CAN_H and CAN_L (R4.8); CAN_H/CAN_L test header (R4.6)
One per bus per zone (R4.1); own 16 MHz crystal (R4.2)

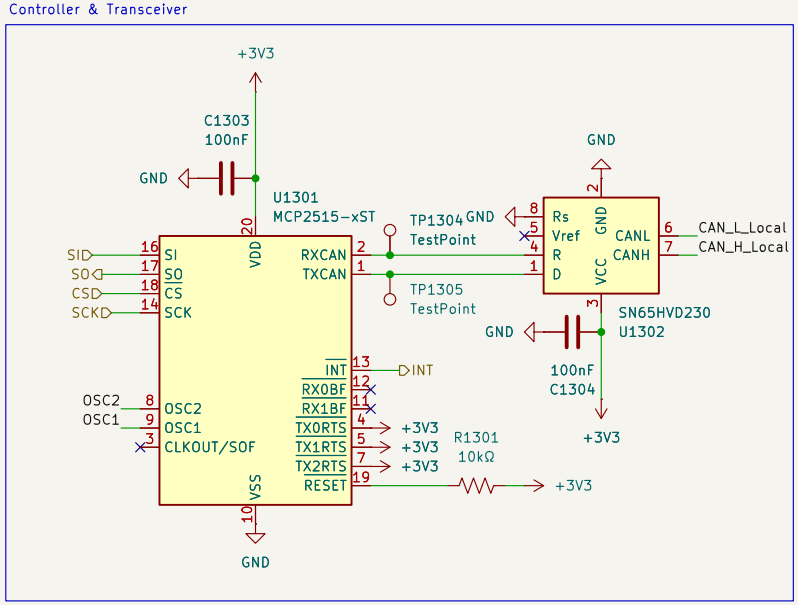
18-649 Lab 4: Controller PCB

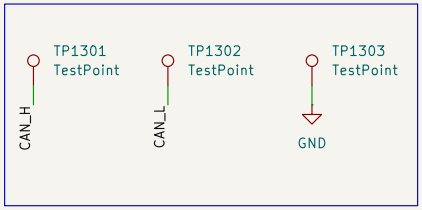
Sheet: /Cockpit/CAN A/
File: can_bus.kicad_sch

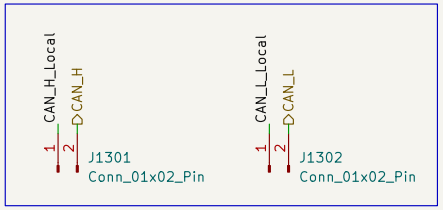
Title: CAN Node: MCP2515 + SN65HVD230 (6 instances)

Size: A4	Date: 2026-10-24	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 12/13









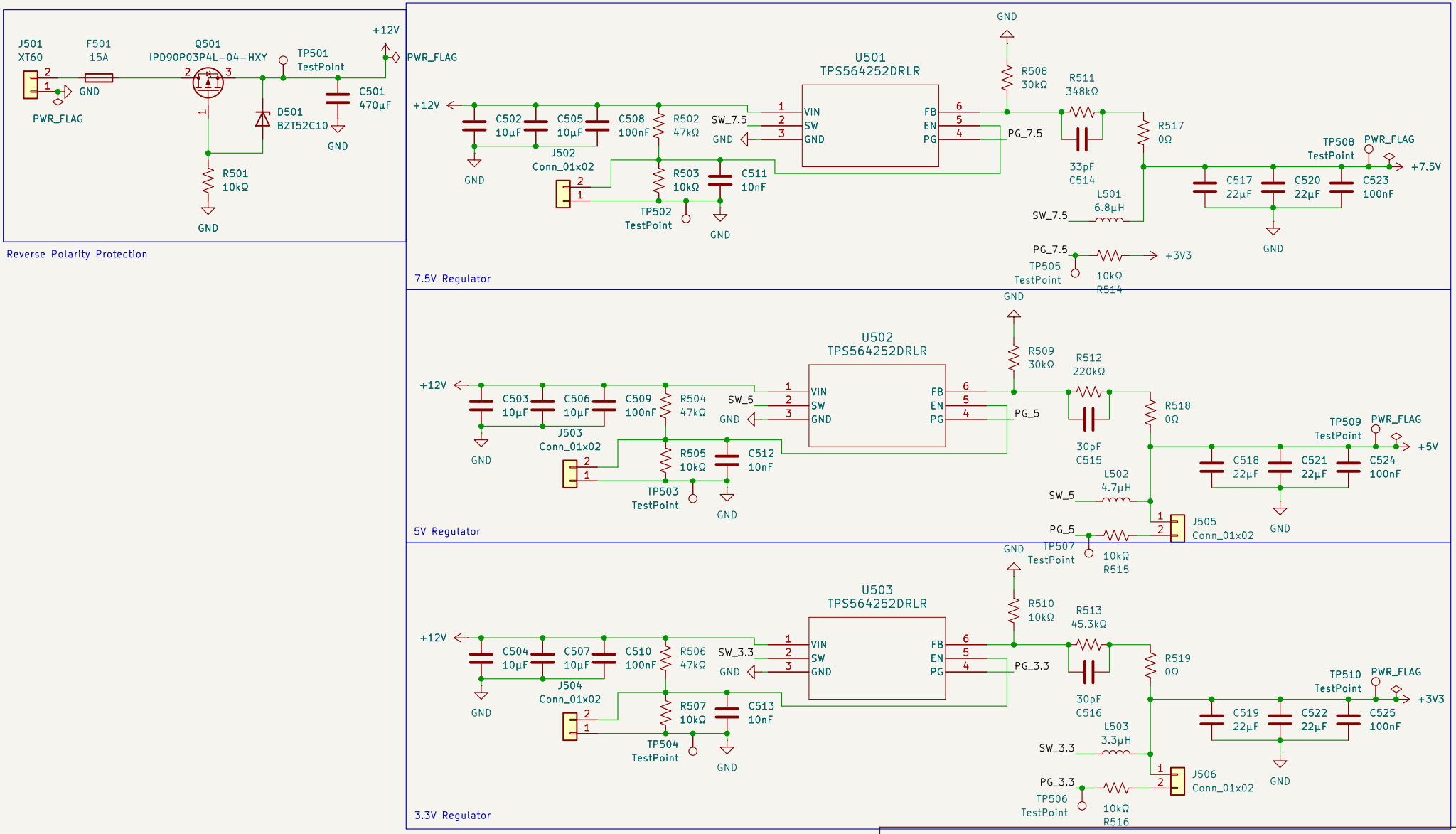
CAN_TX/CAN_RX test points per node
Isolation jumpers break CAN_H and CAN_L (R4.8); CAN_H/CAN_L test header (R4.6)
One per bus per zone (R4.1); own 16 MHz crystal (R4.2)

18-649 Lab 4: Controller PCB

Sheet: /Cockpit/CAN B/
File: can_bus.kicad_sch

Title: CAN Node: MCP2515 + SN65HVD230 (6 instances)

Size: A4	Date: 2026-10-24	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 13/13


J501
XT60
F501
15A
Q501
IPD90P03P4L-04-HXY
TP501
TestPoint
+12V
PWR_FLAG
GND
C501
470µF
D501
BZT52C10
R501
10kΩ
Reverse Polarity Protection
U501
TPS564252DRLR
C502
10µF
C505
10µF
C508
100nF
R502
47kΩ
SW_7.5
J502
Conn_01x02
R503
10kΩ
C511
10nF
TP502
TestPoint
R508
30kΩ
R511
348kΩ
33pF
C514
R517
0Ω
L501
6.8µH
C517
22µF
C520
22µF
C523
100nF
TP508
TestPoint
+7.5V
PG_7.5
TP505
TestPoint
10kΩ
R514
+3V3
7.5V Regulator
U502
TPS564252DRLR
C503
10µF
C506
10µF
C509
100nF
R504
47kΩ
SW_5
J503
Conn_01x02
R505
10kΩ
C512
10nF
TP503
TestPoint
R509
30kΩ
R512
220kΩ
30pF
C515
R518
0Ω
L502
4.7µH
C518
22µF
C521
22µF
C524
100nF
TP509
TestPoint
+5V
PG_5
TP507
TestPoint
10kΩ
R515
J505
Conn_01x02
5V Regulator
U503
TPS564252DRLR
C504
10µF
C507
10µF
C510
100nF
R506
47kΩ
SW_3.3
J504
Conn_01x02
R507
10kΩ
C513
10nF
TP504
TestPoint
R510
10kΩ
R513
45.3kΩ
30pF
C516
R519
0Ω
L503
3.3µH
C519
22µF
C522
22µF
C525
100nF
TP510
TestPoint
+3V3
PG_3.3
TP506
TestPoint
10kΩ
R516
J506
Conn_01x02
3.3V Regulator
EN undervoltage dividers (on 6.8 V / off 5.7 V), PG outputs, enable jumpers
3x TPS564252 buck: +7.5V servo, +5V Pi/logic, +3V3 MCU/CAN
XT60 input, 15 A blade fuse, P-FET reverse polarity protection, 470 uF bulk
18-649 Lab 4: Controller PCB
Sheet: /Power/
File: power.kicad_sch
Title: Power: Battery Input and Regulators
Size: A4
Date: 2026-10-24
Rev: A
KiCad E.D.A. 10.0.7
Id: 5/13